

OpenClaw Servo Control Tutorial

Before starting, please complete the peripheral hardware configuration, API Key configuration, model configuration, and three dialogue scheme configuration tests under the "Preparation Before Use" directory. If you are currently only using TUI/WhatsApp, you can skip the voice interaction configuration. This article assumes the Jetson Nano Docker environment is already prepared and only covers servo-related content.

1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capability on Jetson Nano through the servo peripheral. After completing this tutorial, readers will be able to:

- Understand the basic approach for OpenClaw on Jetson to control servos
- Control servos via WhatsApp, TUI, and voice three methods
- Complete cases including single servo control, multi-servo synchronization, servo Q&A, table tennis simulation, and pointer simulation

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw expansion board
- Servo [1 piece / 2 pieces]
- [Optional] AI voice interaction module + speaker

Note:

- Servo angle range is 0-180 degrees.
- Servo numbers use 1-2.

3. Software Preparation

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Jetson-side entry points used in this tutorial:

```
/root/.openclaw/openclaw.json  
/root/.openclaw/workspace/skills/jetson-hardware-control/SKILL.md  
Corresponding atomic tools (servo/rgb/dht/oled/camera/music)  
/root/yahboom_ws/src/largemodel/main.py
```

4. Test the Servo

Before starting this tutorial, please complete one `openclaw_tui` basic dialogue self-check. After confirming OpenClaw can respond normally, proceed to servo-specific testing.

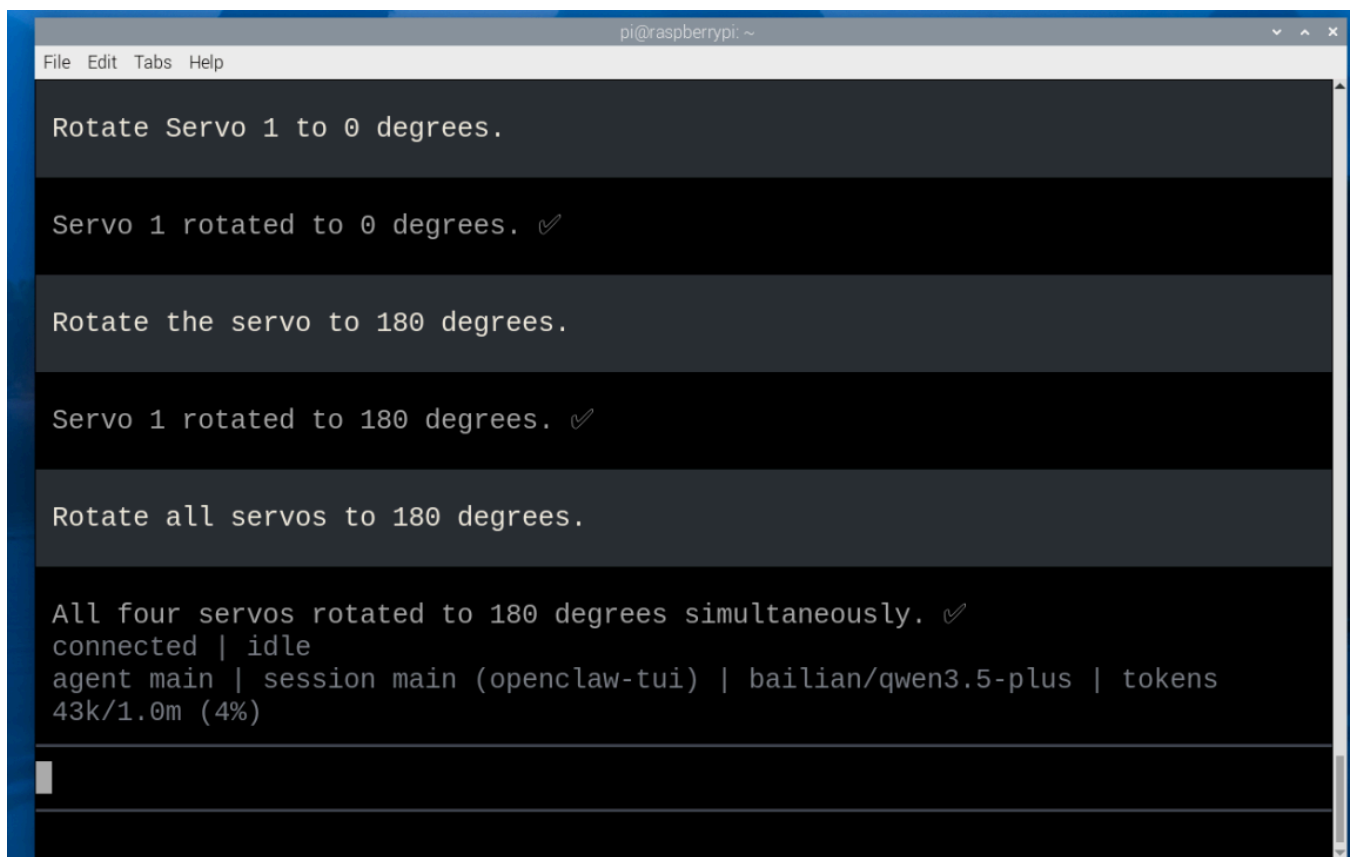
Enter TUI by typing the following command in the terminal:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔊 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

After ensuring the hardware is properly connected, you can directly input in the TUI:

- Rotate servo 1 to 0 degrees
- Rotate servo 1 to 90 degrees
- Rotate all servos to 180 degrees



```
pi@raspberrypi: ~
File Edit Tabs Help

Rotate Servo 1 to 0 degrees.

Servo 1 rotated to 0 degrees. ✓

Rotate the servo to 180 degrees.

Servo 1 rotated to 180 degrees. ✓

Rotate all servos to 180 degrees.

All four servos rotated to 180 degrees simultaneously. ✓
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
43k/1.0m (4%)
```

If you need to bypass the model and directly test hardware commands, execute the following in the container:

```
servo call-tool Servo1 --action set_angle --angle 0
servo call-tool Servo1 --action set_angle --angle 90
servo call-tool Servo1 --action set_angle --angle 180
```

5. Three Schemes for Dialogue with OpenClaw

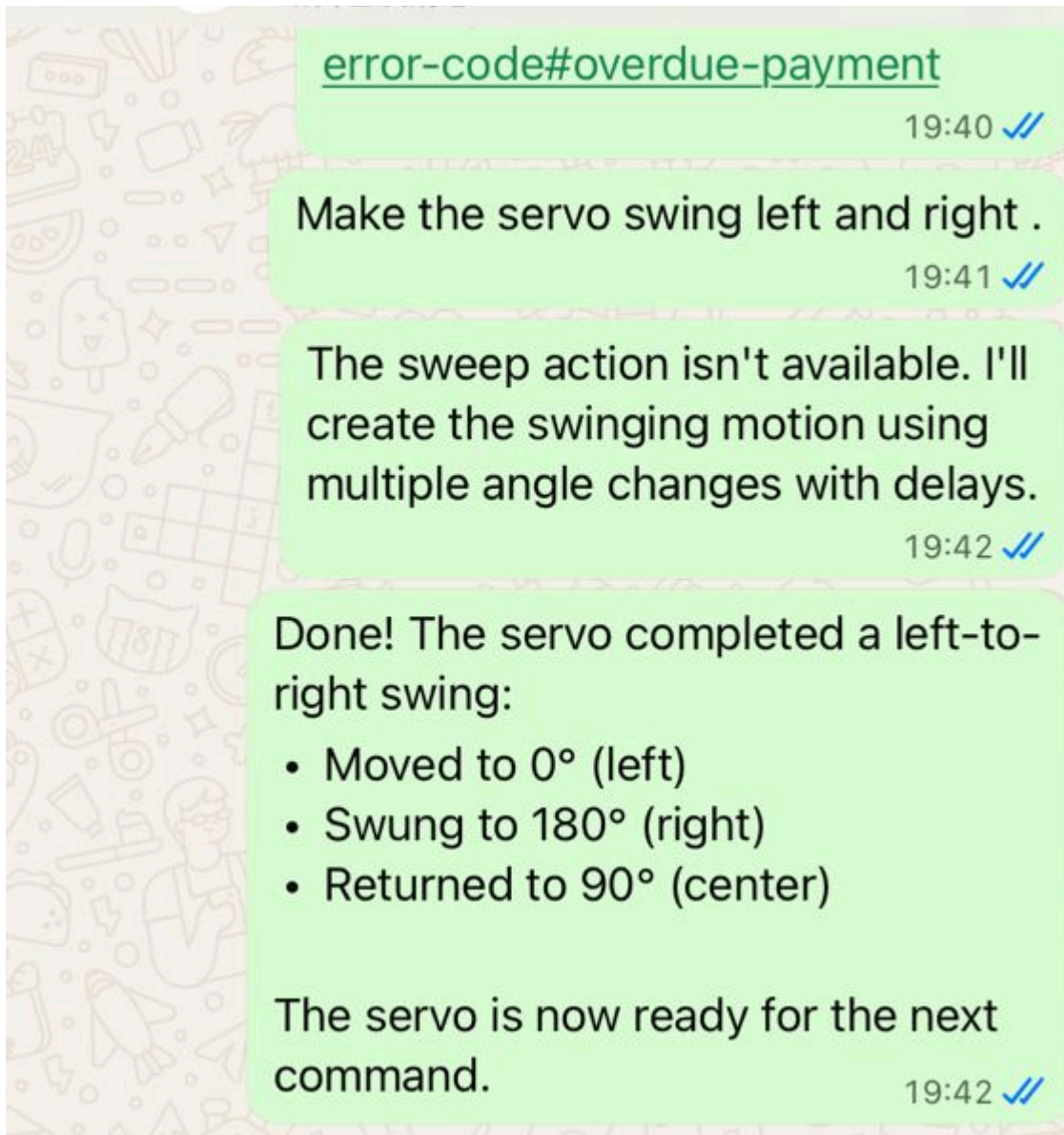
Scheme	Suitable Scenario	Advantages	Suggested Positioning
WhatsApp	Remote control, classroom demonstration	Convenient on mobile, suitable for multi-person experience	Demonstrate remote natural language control
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First get the control link working
Voice Interaction	Voice wake-up, complete AI experience	Closest to real dialogue experience	As the final demonstration solution

5.1 WhatsApp Scheme

For WhatsApp access, QR code creation, and gateway startup steps, please refer to "Three Dialogue Scheme Configuration Tests". This section only retains dialogue examples suitable for the servo tutorial.

You can have direct conversations, for example:

- Rotate servo 1 to 0 degrees
- Rotate the servo to 90 degrees
- Make the servo swing left and right



5.2 TUI Scheme

If TUI self-check is already completed, directly input in the container:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔧 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

It is recommended to first test single servo, then test two servos:

- Turn servo 1 to 90 degrees
- Turn servo 2 to 45 degrees

5.3 Voice Interaction Scheme

Requires optional AI voice interaction module and speaker.

Execute the following command when openclaw gateway is running:

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

Suitable for directly demonstrating servo voice commands, for example:

- Rotate the servo to 90 degrees
- Make the servo nod
- Make the servo swing left and right

```
2026-05-14 14:48:20 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6060 ms 语音, 正在转写

[你] Make the servo swing left and right.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me make the servo swing using individual angle commands:
Done!
        The servo swung left to right and back to center. 🤖

The motion
        sequence was:
- Center (90°) → Left (0°) → Right (180°) → Center
  (90°)
[信息] 回复已裁剪为播报文本: 202 -> 116 字
```

If voice has been recognized as text but the servo doesn't move, first check whether `openclaw_bridge` is forwarding the voice text to OpenClaw, and whether OpenClaw generated a servo action.

6. Comprehensive Cases

The following cases can be executed in TUI, WhatsApp, or voice. During the debugging phase, it is recommended to use TUI first.

6.1 Case 1: Servo Dancing

Experiment Objective

Make the servo move continuously through a set of preset angles, forming a "dancing" rhythm effect.

Example Commands

- Make the servo dance
- Make the servo swing left and right quickly

6.2 Case 2: Yes/No Q&A

Experiment Objective

Make the servo express "yes" or "no" through actions, for example, nodding represents "yes", shaking head represents "no".

Example Commands

- Does the sun rise from the east? If so, make the servo nod
- Is $1+1=3$ correct? If not, make the servo shake its head

6.3 Case 3: Table Tennis Simulation

Experiment Objective

Make the servo continuously move left and right, simulating a table tennis ball bouncing back and forth between two sides of the table.

Example Commands

- Make the servo simulate table tennis movement back and forth
- Swing left and right 20 times, faster

6.4 Case 4: Pointer Simulation

Experiment Objective

Treat the servo as a rotatable pointer to indicate values, directions, or status.

Example Commands

- Turn the pointer to 30°
- Simulate a gauge, current value is 75

6.5 Case 5: Multi-Servo Control

Experiment Objective

Make two servos simultaneously turn to target angles to verify multi-servo control capability.

Example Commands

- Turn servo 1 and 2 to 30 and 60 degrees respectively
- Turn both servos to 180 degrees, then light the RGB red

7. Common Problems

- Servo has no action: First execute the hardware command test to confirm whether the atomic tool is working normally.
- Servo only vibrates but doesn't rotate: First check the power supply; when multiple servos move simultaneously, current insufficiency is more likely.
- Controlling the wrong servo: Confirm servo numbers are `1, 2`; in natural language, it is best to explicitly say "servo 1, servo 2".
- Abnormal angles: Confirm the angle is between `0-180` degrees; do not input negative numbers or angles exceeding 180.